Syllabus **Algebra 1**

Algebra 1 covers algebra topics such as exponents and radicals, solving linear equations, linear and nonlinear inequalities, ratio and proportion, solving systems of linear equations, solving quadratic equations, complex numbers, functions, exponential and logarithmic functions, graphing equations and inequalities of varying types, and sequences and series.

This course is specifically designed for high-performing students and draws material from many programs for top middle and high school students in the country. Our philosophy is that students develop more by learning to solve problems they haven't seen before, as opposed to offering repeated drills that students can memorize their way through. In this way, our classes are structured much more like courses at top-tier colleges.

Books: *Introduction to Algebra* by Richard Rusczyk

Time Commitment: This 36-week course includes 105 minutes in-class and 1-3 hours of homework each week, corresponding to a full year course.

Content:

Week	Topic
1	Exponents and Radicals
2	Variable Expressions
3	Linear Equations and Word Problems
4	Multi-Variable Expressions
5	Introduction to Systems
6	Solving Systems of Equations
7	Ratios and Conversions
8	Percents
9	Proportions and Rates
10	Connecting Graphs and Equations
11	Graphing Lines, Part 1
12	**Midterm 1**
13	Graphing Lines, Part 2
14	Inequalities
15	Factoring Monic Quadratics
16	Factoring Non-Monic Quadratics
17	Vieta's Formulas
18	Complex Numbers

Week	Topic
19	Completing the Square
20	The Quadratic Formula
21	Graphing Quadratics, Part 1
22	Graphing Quadratics, Part 2
23	Nonlinear Inequalities
24	**Midterm 2**
25	Introduction to Functions
26	Composition and Inverse
27	Graphing Functions
28	Transformations of Functions
29	Polynomial Functions
30	Exponential Functions
31	Logarithmic Functions
32	Special Functions
33	Sequences and Series, Part 1
34	Sequences and Series, Part 2
35	Sequences and Series, Part 3
36	**Final Exam**

Common Core State Standards:

Domain	Subdomain	Standards
Number and Quantity	The Real Number System	1, 2
	Quantities	1, 2
	The Complex Number System	1, 2, 3, 7, 8, 9
Algebra	Seeing Structure in Expressions	1ab, 2, 3abc, 4
	Arithmetic with Polynomials & Rational Expressions	1, 4, 7
	Creating Equations	1, 2, 3, 4
	Reasoning with Equations and Inequalities	1, 2, 3, 4ab, 6, 7, 10, 11, 12
Functions	Interpreting Functions	1, 2, 3, 4, 5, 7abcde, 8ab, 9
	Building Functions	1abc, 2, 3, 4abc, 5
	Linear, Quadratic, & Exponential Models	1abc, 2, 4, 5
Geometry	Expressing Geometric Properties with Equations	2, 5, 6

Sample Problems:

- Find all pairs of real numbers (x, y) such that $x + y = 6$ and $x^2 + y^2 = 28$.
- Suppose R is $(3, 5)$ and S is $(8, -3)$. Find each point on the line through R and S that is three times as far from R as it is from S.
- A line L has a slope of -2 and passes through the point $(r, -3)$. A second line, K, is perpendicular to L at (a, b) and passes through the point $(6, r)$. Find a in terms of r.
- Find all complex numbers $a + bi$ such that $\overline{a + bi} = (a + bi)^2$.
- The function $f(x) = mx + b$ is such that f and f^{-1} are the same function. Find all possible values of m.
- What is the value of the infinite sum $27 + 18 + 12 + 8 + \cdots$?